# Practice workbook · Caesar cipher

Encrypting/decrypting with a Caesar shift, finding the shift, combining shifts, spotting wrap-around errors and explaining weakness.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-18

# 1. Question 4 · part a.i · 2 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part a.i only.

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**4** Data is encoded for encryption and for identifying records in databases.

(a) A Caesar cipher is one method of encryption.

(i) Complete the table to show the result of applying a Caesar cipher.

**(2)**

| Plaintext | Shift | Ciphertext |
|---|---|---|
| PIXEL | −4 | |
| CLOUD | | FORXG |

(ii) Plaintext is encrypted using a Caesar cipher.

Compare the ciphertexts produced for the plaintext **GOLD** by:

- a shift of −6 followed by a shift of +8
- a single shift of +2.

**(2)**

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(iii) Ann has written a Caesar cipher algorithm.

When the algorithm encrypts the plaintext **BYTE** with a shift of +5 the result is **GVYJ**.

This is incorrect.

Explain the error.

**(2)**

## 1. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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## 2. Question 3 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

**3** Several encryption algorithms have been developed.

(a) Ahmed has started to create a Caesar cipher encryption program.

Caesar cipher encryption works by giving a number value to a key.

Each letter in a plaintext message is replaced by a new ciphertext letter using the key.

For example, as shown in the table, a key value of +2 would change the plaintext message **encrypt** to the ciphertext message **gpetarv**.

| **Plaintext** | e | n | c | r | y | p | t |
|---|---|---|---|---|---|---|---|
| **Ciphertext** | g | p | e | t | a | r | v |

Ahmed's program already converts from plaintext to ciphertext.

He wants to extend the program to:

- accept the input of a plaintext message
- accept the input of the key, which must be a positive integer between 1 and 25
- write the generated ciphertext to a text file named **Cipher.txt**
- display the ciphertext.

Open **Q03a** in the code editor.

Complete the program code.

You must use the structure given in the file **Q03a** to complete the program.

Do not add any further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

**(8)**

(b) Explain **one** reason why the Caesar cipher is a weak encryption algorithm.

**(2)**

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## 2. Question 3 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

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(c) The Vigenere cipher encrypts plaintext using a series of Caesar ciphers, based on the letters of a keyword and a Vigenere square.

**Figure 1** shows a Vigenere square.

| | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| a | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
| b | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a |
| c | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b |
| d | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c |
| e | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d |
| f | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e |
| g | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f |
| h | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g |
| i | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h |
| j | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i |
| k | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j |
| l | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k |
| m | m | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l |
| n | n | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m |
| o | o | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n |
| p | p | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o |
| q | q | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p |
| r | r | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q |
| s | s | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r |
| t | t | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s |
| u | u | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t |
| v | v | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u |
| w | w | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v |
| x | x | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w |
| y | y | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x |
| z | z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y |

**Figure 1**

Complete the table to perform a Vigenere decryption of the ciphertext message **flmktrwhee** using the keyword **thirty**.

**(4)**

| Encrypted letter | f | l | m | k | t | r | w | h | e | e |
|---|---|---|---|---|---|---|---|---|---|---|
| Keyword letter | | | | | | | | | | |
| Decrypted letter | | | | | | | | | | |

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## 2. Question 3 · part b · 2 marks · Working and answer

Working and answer

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## 3. Question 4 · part a.ii · 2 marks

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Further practice · Attempt part a.ii only.

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(i) Complete the table to show the result of applying a Caesar cipher.

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(ii) Plaintext is encrypted using a Caesar cipher.

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When the algorithm encrypts the plaintext **BYTE** with a shift of +5 the result is **GVYJ**.

This is incorrect.

Explain the error.

**(2)**

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## 3. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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# 4. Question 4 · part a.iii · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a.iii only.

**4** Data is encoded for encryption and for identifying records in databases.

(a) A Caesar cipher is one method of encryption.

(i) Complete the table to show the result of applying a Caesar cipher.

**(2)**

| Plaintext | Shift | Ciphertext |
|---|---|---|
| PIXEL | −4 | |
| CLOUD | | FORXG |

(ii) Plaintext is encrypted using a Caesar cipher.

Compare the ciphertexts produced for the plaintext **GOLD** by:

- a shift of −6 followed by a shift of +8
- a single shift of +2.

**(2)**

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When the algorithm encrypts the plaintext **BYTE** with a shift of +5 the result is **GVYJ**.

This is incorrect.

Explain the error.

**(2)**

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# 4. Question 4 · part a.iii · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a.iii only.

(b) A program is needed to create a key for a database.

A user enters a two-letter word, a whole number and a decimal number.

The program must ensure the word is only two characters long.

The program must display an error message when the word is not two characters long.

A key is generated from the whole number, the reversed word and the whole number part of the decimal number.

**Figure 4** shows the input values and a valid key.

```
Enter a word: at
Enter a whole number: 12
Enter a decimal number: 7.89
12ta7
```

**Figure 4**

Open **Q04b** in the code editor.

Write the program.

Do not add any further functionality.

Save your code as **Q04bFINISHED** with the correct file extension for the programming language.

(8)

**(Total for Question 4 = 14 marks)**

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## 4. Question 4 · part a.iii · 2 marks · Working and answer

Working and answer

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